Heart Disease Risk Analysis Prediction

From Raw Data to Clinical Insights: A Multi-Stage Data Science Approach

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1. Introduction:

Heart diseases are the leading cause of death globally, taking an estimated 17.9 million lives each year. These disorders affect the heart and blood vessels and include conditions such as coronary heart disease and heart failure. More than four out of five CVD deaths are due to heart attacks and strokes, and one-third of these deaths occur prematurely in people under 70 years of age.

Most Heart diseases can be prevented by addressing behavioral risk factors such as unhealthy diet, obesity, and physical inactivity. Therefore, early detection and management of risk factors—ranging from clinical symptoms like chest pain to lifestyle indicators like smoking—are crucial in reducing mortality rates.

2. Project Objectives:

The primary goal of this project is to bridge the gap between **clinical symptoms** and **predictive analytics** to enhance early heart disease detection. Using a comprehensive dataset of 15+ features, this analysis seeks to:

- **Identify High-Risk Profiles:** Analyze the correlation between 17 binary variables (symptoms and risk factors) and the likelihood of heart disease.
- **Analyze Risk of heart disease by the Predictors:** Measure the risk by **Age** and its interaction with lifestyle choices (e.g., smoking and obesity).
- **Develop a Predictive Insight Model:** Leverage machine learning techniques to calculate a **Risk Probability Score**, allowing for a data-driven approach to patient screening.
- **Propose Proactive Interventions:** Provide actionable recommendations for healthcare providers to prioritize patients based on the "Feature Importance" identified through the analysis.

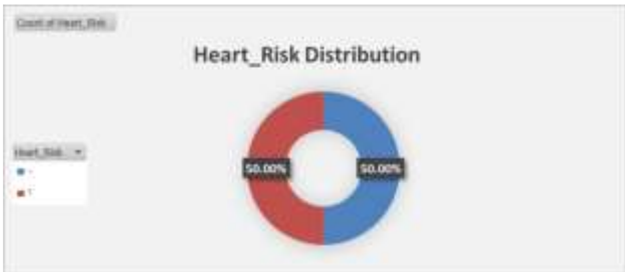
3. Data Description:

The dataset was imported from kaggle , It contains 70,000 records representing unique patient profiles : 18 features covering clinical symptoms, lifestyle habits, and demographic data.

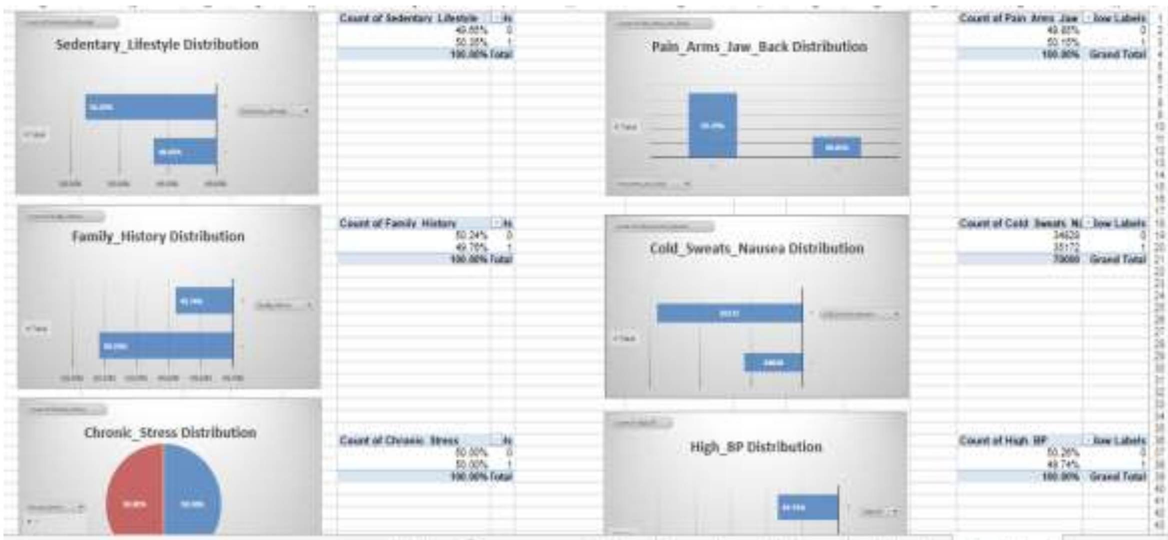
Column Name	Description	Data Type	Example Values
chest_pain	Presence of chest pain (Yes/No)	Binary	0 (No), 1 (Yes)
shortness_of_breath	Difficulty breathing (Yes/No)	Binary	0 (No), 1 (Yes)
fatigue	Persistent tiredness without an obvious cause (Yes/No)	Binary	0 (No), 1 (Yes)
palpitations	Irregular or rapid heartbeat (Yes/No)	Binary	0 (No), 1 (Yes)
dizziness	Episodes of lightheadedness or fainting (Yes/No)	Binary	0 (No), 1 (Yes)
swelling	Swelling in legs/ankles due to fluid retention (Yes/No)	Binary	0 (No), 1 (Yes)
radiating_pain	Pain radiating to arms, jaw, neck, or back (Yes/No)	Binary	0 (No), 1 (Yes)
cold_sweats	Cold sweats and nausea (Yes/No)	Binary	0 (No), 1 (Yes)
age	Patient's age in years	Continuous	min : 20 - max : 84
hypertension	History of high blood pressure (Yes/No)	Binary	0 (No), 1 (Yes)
cholesterol_high	Elevated cholesterol levels (Yes/No)	Binary	0 (No), 1 (Yes)
diabetes	Diagnosis of diabetes (Yes/No)	Binary	0 (No), 1 (Yes)
Gender	Patient's biological sex	Binary	0 (Female), 1 (Male)
smoker	Smoking history (Yes/No)	Binary	0 (No), 1 (Yes)
obesity	Obesity status (Yes/No)	Binary	0 (No), 1 (Yes)
family_history	Family history of heart disease (Yes/No)	Binary	0 (No), 1 (Yes)
risk_label	Risk of heart disease (Low Risk , High Risk)	Binary	0(Low risk) , 1(High risk)

Variables' Distribution :

Target Variable : Risk of Heart Disease



Subset of Explanatory Variables' Distribution



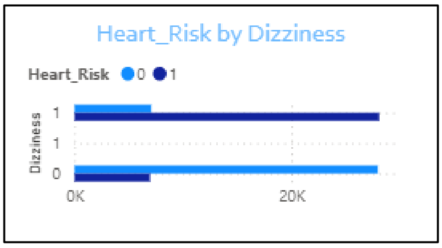
4. Prevalence Analysis:

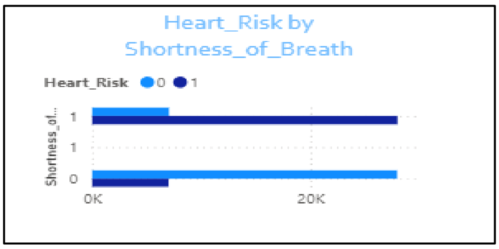
Key Insights

The analysis shows that age is the most significant factor influencing heart disease risk. We can clearly see that people with heart disease risk (1) have a higher average age (64 years) compared to those without risk (44 years).

*There are Symptoms Correlated by Heart Risk

Chest Pain Fatigue





Dizziness Shortness of Breath

Pain in arms/jaw/back Cold sweats & nausea

The disease is clearly manifested through symptoms, and the presence of more than one symptom provides a serious risk

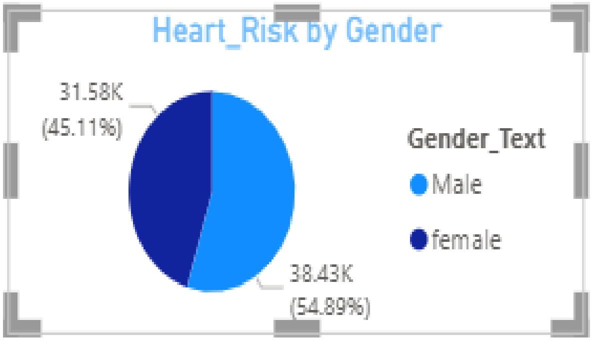
There are Risk Factors Correlated by Heart Risk

High_BP High_Cholesterol

Smoking Diabetes

Obesity Sedentary_Lifestyle

Family_History Chronic_Stres



Gender Its impact is weak compared to other factors but Age shows that age is the most significant factor influencing heart disease risk.

Challenges

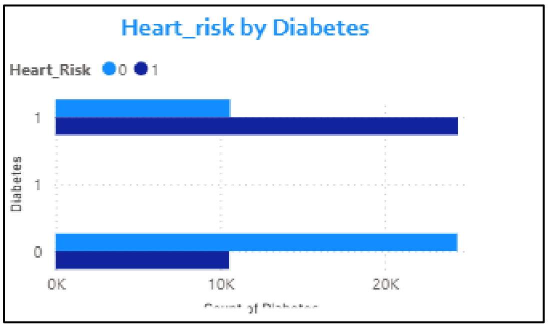
1. Aging Population

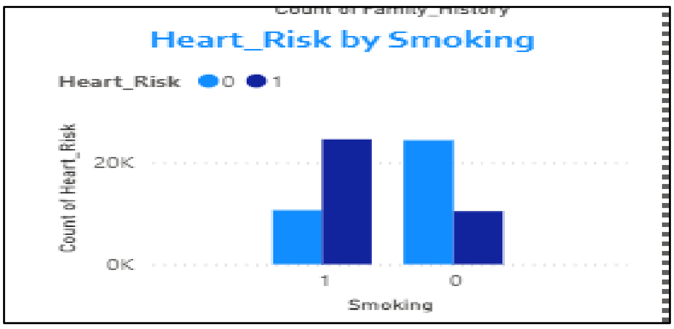
* Increased risk with age

* Higher vulnerability among older adults

"Heart disease risk increases significantly with age."

2. Unhealthy Lifestyle





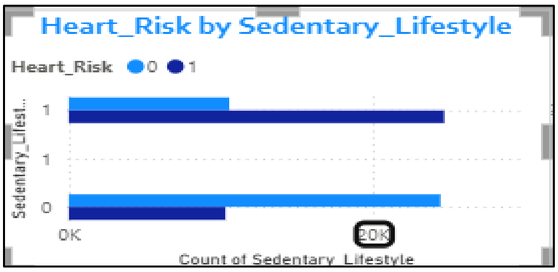
* High rates of smoking

* Poor diet (fast food, high fat intake)

* Lack of physical activity

"Unhealthy lifestyle habits are major contributors to heart disease."

3. Sedentary Behavior

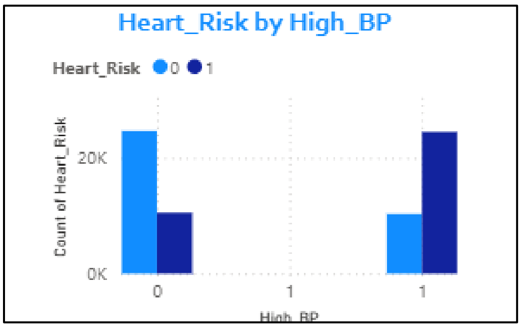


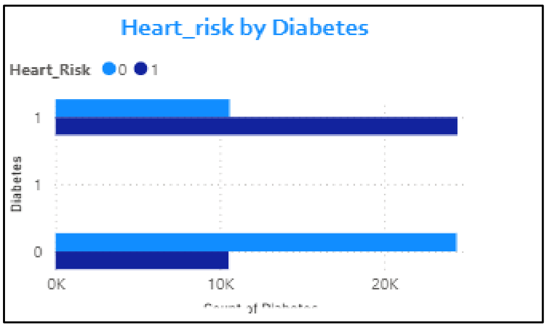
* Long hours of sitting

* Low daily physical movement

"A sedentary lifestyle increases the likelihood of heart disease."

4. Chronic Diseases Prevalence





* High blood pressure

* Diabetes

* High cholesterol

“Chronic conditions significantly increase heart disease risk.

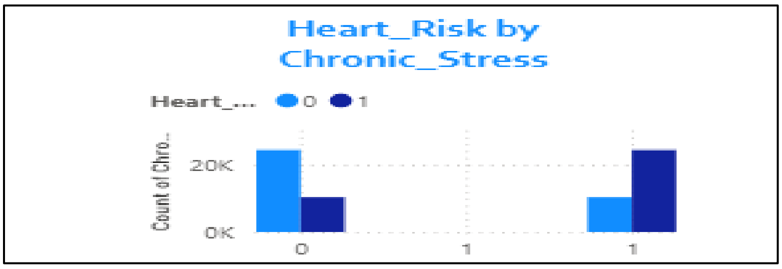
5. Late Detection

* Symptoms appear at advanced stages

* Lack of regular health checkups

“Late detection reduces the chances of effective treatment.”

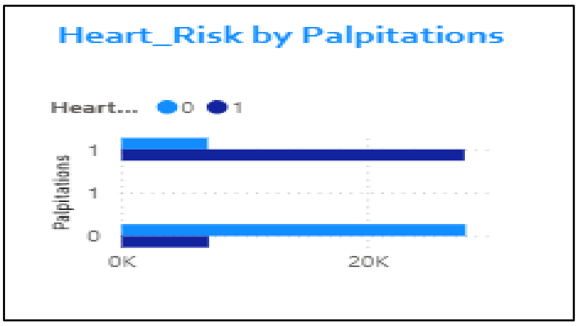
6. Stress and Mental Pressure



* Chronic stress

* Poor work-life balance

“Long-term stress negatively impacts heart health.”



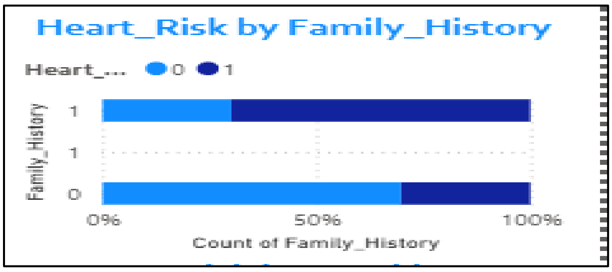
7. Lack of Awareness

* Limited knowledge about symptoms

* Ignoring early warning signs

“Lack of awareness delays diagnosis and treatment.”

8. Genetic Factors



* Family history of heart disease

"Genetic predisposition plays a role but is often unavoidable."

Elderly individuals

Patients with diabetes or hypertension

Smokers

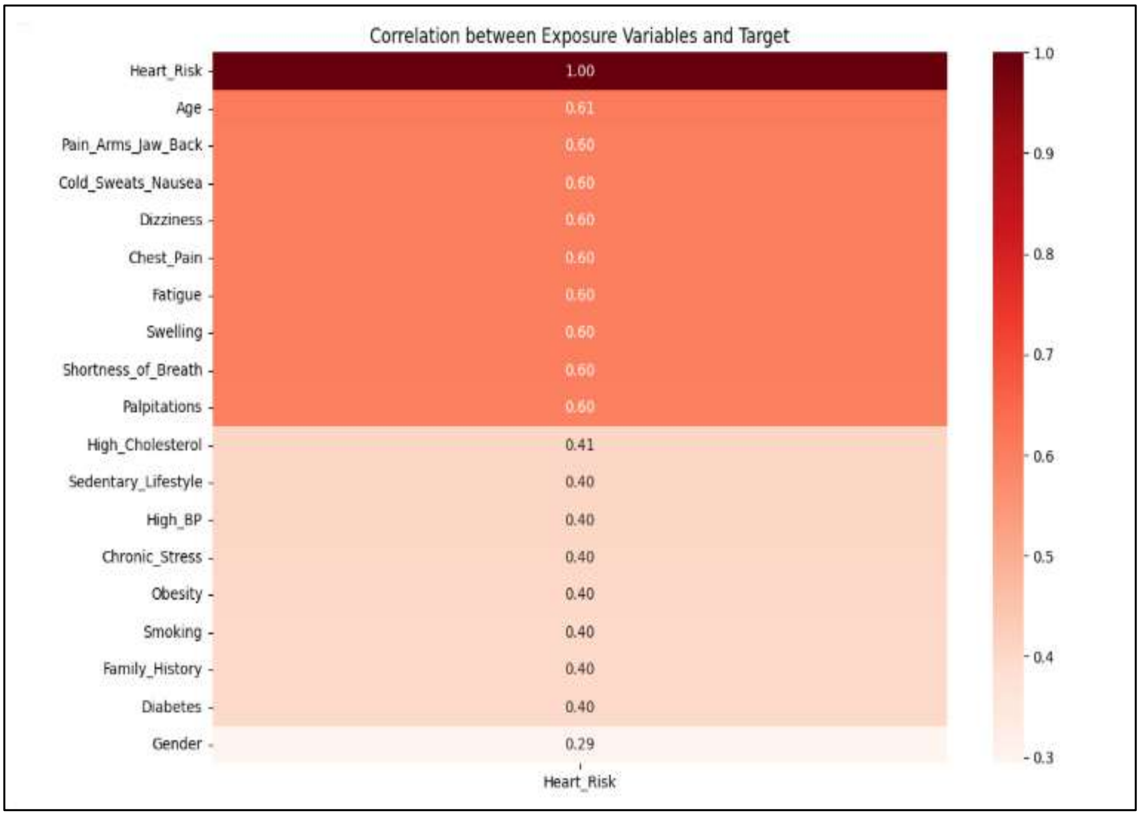
"Focusing on high-risk groups improves prevention strategies."

Conclusion

"Prevention, early detection, and lifestyle changes are the key strategies to reduce heart disease risk."

5. Exposures Impact and Risk Assessment :

- ### Correlation Heat Map :



The correlation heat map provides an overview of the relationships between the variables and It showed that :

- **Age** has **the strongest positive correlation** with the risk of heart disease, revealing that it is the foundational factor underlying the disease risk pattern .

- **Acute Symptoms :** Such as Chest Pain , Swelling , Fatigue & Shortness of breath show **a strong positive** correlation with heart disease risk .

- **Lifestyle & Chronic Factors :** Such as Smoking , Sedentary , High Blood Pressure & Cholesterol have **a moderate positive** correlation ,suggesting that they have **a cumulative impact** rather than immediate.

To enhance the study analysis , we implemented a **Logistic Regression Model** as a diagnostic tool rather than data description , the allowed us to :

- **Quantify Risk :** Calculate the exact risk probability that an individual might have a heart disease .
- **Measure Impact :** Extract the strength of effect for each exposure and Identify which of them acts as a **strong red flag .**

Key Findings :

The model showed that life style factors combined with clinical symptoms provide a **99% accurate prediction ,** this will enable us to categorize patients into High , Moderate and Low risk groups for medical resource allocation .

Exposures Impact :

Variable	Impact	Interpretition
Fatigue	**18.8**	Fatigue **increases** the risk of heart disease by **18.8 times**.
Swelling	**18.5**	Swelling **increases** the risk of heart disease by **18.5** times.
Cold_Sweats_Nausea	**17.6**	Individuals with cold sweats Nausa have **70% higher risk** of heart disease than those who don't.
Chest_Pain	**15.7**	Chest pain **increases** the risk of heart disease by **15 times.**
Pain_Arms_Jaw_Back	**15.6**	Occurance of Arms, Jaw and Back pain **increase** the odds of heart disease by **15 times** .
Palpitations	**15.5**	Palpitations **increase** the odds of heart disease by **15.5** times .
Dizziness	**14.9**	Dizziness **increase** the odds of heart disease by **14.9** times .
Shortness_of_Breath	**14.6**	People suffering shortness of breath have a **40% higher risk** that those who are not .
Chronic_Stress	**6.1**	Chronic stressed individuals have **6 times the risk** of heart disease compared to the Non-stressed .
High_Cholesterol	**5.8**	High Cholestrol **increases** the odds of heart disease by **5.8 times** .
Smoking	**5.8**	Smoking **increases** the odds of heart disease by **5.8 times** .
Family_History	**5.7**	Individuals with family history of heart disease have **5.7 times** the risk of it among those who are not .
Sedentary_Lifestyle	**5.6**	Sedentarism **increases** the risk of heart disease by **400 %**.
Obesity	**5.5**	Obesity **increases** the odds of heart disease by **5.5 times** .
Diabetes	**5.4**	Diabetes **increases** the odds of heart disease by **5.4 times** .
High_BP	**5.4**	High Blood Pressure **increases** the odds of heart disease by **5.4 times** .
Gender	**3.5**	Risk of heart disease among males **have 3.5 time** the risk of heart diseas among females .
Age	**1.1**	**Each year increased in age** by year **icreases** the risk by **10% .**

The table shows that the effect of exposures is classified into two levels :

Level 1 : Immediate Red Flags :

Symptoms as Fatigue , Shortness of breath Arms , Neck and Jaw Pains are **the most critical indicators** , as each of them independently increases the risk of heart disease by **over 14 times** .

These are acute indicators requiring **immediate medical intention** .

Level 2 : Chronic Risk Drivers :

Lifestyle factors (smoking , Sedentry lifestyle) & and clinical markers like cholesterol & high blood pressure that show significant impact that each of them increases the risk of heart disease by **3 to 6 times**.

These factors as **long-Term Contributors** to the development of the heart disease .

Age Effect :

Age as a continuous variable , the model shows it acts as the environment in which other symptoms become more critical.

- ## Final Risk Profile :

With probability of heart disease risk obtained from the model , Patients were classified into three categories based on Risk Probability , and a final risk profile that defines the highest common exposures among each group .

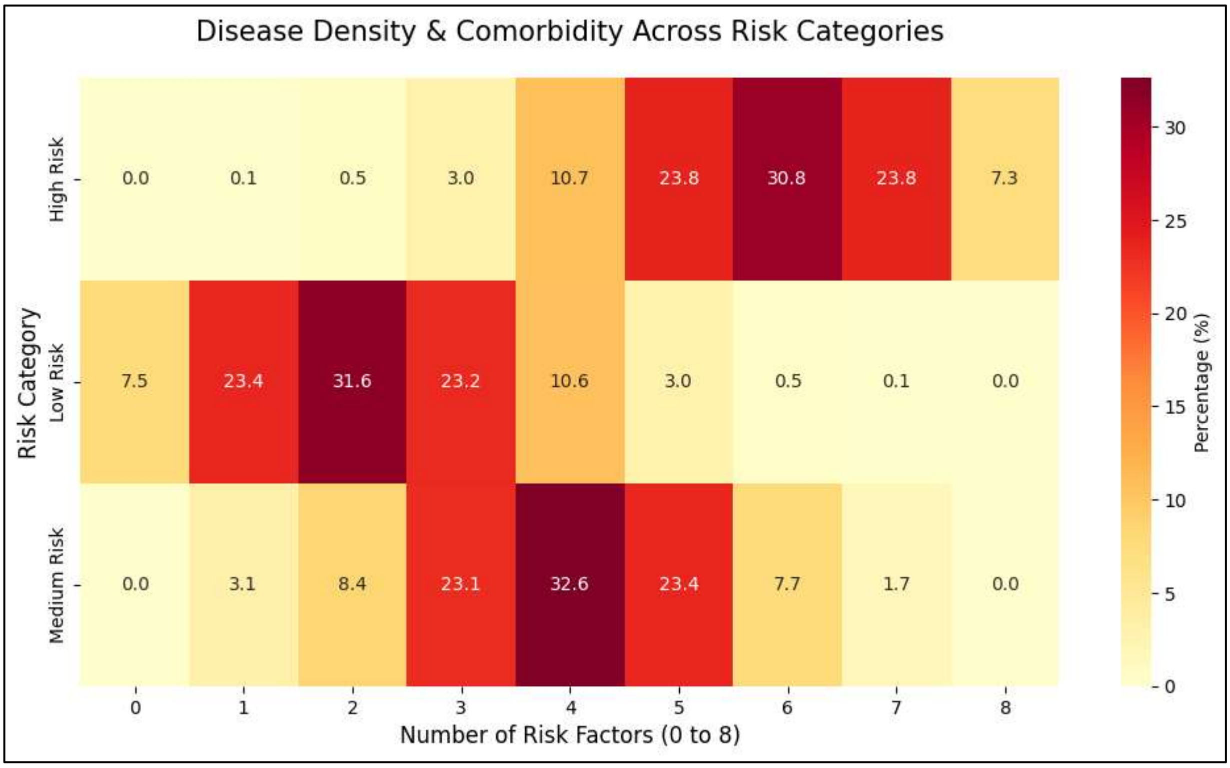
The following tables revel the Age's Range and Mean , Contribution of the following exposures .

Risk_Category	Age_Min	Age_Max	Age_Mean	High_BP	High_Cholesterol	Smoking	Diabetes	Obesity	Chest_Pain	Chronic_Stress
High Risk	23	84	64.47	70%	70%	70%	70%	70%	80%	70%
Medium Risk	20	84	56.61	51%	50%	49%	51%	52%	48%	44%
Low Risk	20	82	44.4	29%	29%	30%	30%	30%	20%	30%

Following The mean age , Prevalence of the previous factors from low to high risk of heart disease show that age and the Comorbidities show a linear progression with risk levels.

- ## Risk Profile Heat Map :

To Explore the density of factors in each Risk Category and explore how cumulative factors lead to higher risk of disease.



The map shows that :

High Risk Category : about 78% of them suffer from 5 to 7 exposures, with average age of 64 years , the old age is when all factors are combined to the highest risk of heart disease occurrence .

Medium Risk Category : about 79 % of them have from 3 to 5 diseases which represents a Turning point that another additional exposure will push them to the High Risk zone .

Low Risk Category : The safe Zone The majority of people in this category have from 1 to 3 exposures .

6. Conclusions and Recommendations :

- **Final Conclusion**

The analysis transcends traditional risk assessment by demonstrating that heart disease risk is not driven by isolated factors, but by their **cumulative density**. Using our Logistic Regression model, we identified a **"Critical Threshold" at 4 risk factors**; this is the tipping point where a patient transitions from manageable risk to a high-probability danger zone.

Key Insight: While age is a factor, the real driver is **Comorbidity Progression**. Our findings show that **78% of High-Risk individuals** carry between **5 to 7 concurrent factors**, creating a "synergistic" effect that exponentially increases risk.

Recommendation: Clinical interventions must shift from treating single symptoms to **integrated management strategies**. By targeting the "Turning Point" (patients with 3–4 factors), healthcare providers can effectively prevent the transition into the High-Risk category, potentially saving lives through proactive, data-driven prevention.

- **Recommendations :**

"Heart disease is driven by a combination of lifestyle, medical, and demographic challenges."

1. Lifestyle Improvement

Regular physical activity (at least 30 minutes daily)

Reduce or quit smoking

Reduce fatty and fast foods

Maintain a healthy weight

"Adopting a healthy lifestyle is essential to reduce heart disease risk.

2. Early Detection

Regular checkups for:

Blood pressure

Cholesterol levels

Blood sugar

Continuous medical follow-up

"Regular checkups help detect heart disease at an early stage."

3. Chronic Disease Management

Control and monito

High Blood Pressure

Diabetes

High Cholesterol

Follow prescribed treatments

"Managing chronic diseases significantly reduces complications and risk."

4. Stress Management

Improve sleep quality

Practice relaxation techniques

Reduce daily stress

"Reducing stress plays an important role in preventing heart disease."

5. Physical Activity

Avoid long periods of sitting

Stay active through daily movement or exercise

"An active lifestyle helps improve heart health."

6. Awareness

Educate people about key symptoms:

*Chest pain * Shortness of breath

Encourage early medical consultation

"Awareness can save lives by encouraging early medical intervention."

7. Target High-Risk Groups

Elderly individuals

Patients with diabetes or hypertension

Smokers

"Focusing on high-risk groups improves prevention strategies."

"Prevention, early detection, and lifestyle changes are the key strategies to reduce heart disease risk."

Appendix : Binary Logistic Regression Model :

1. Why Binary Logistic Regression?

Binary Logistic Regression was selected because the dependent variable is **dichotomous** (e.g., Presence vs. Absence of Disease). Unlike Linear Regression, which can predict values outside the $[0, 1]$ range, Logistic Regression uses the **logit link function** to constrain the output to a probability. It is particularly effective for large datasets (e.g., $N = 70,000$) where the goal is to determine the **Odds Ratio (OR)** of specific risk factors.

2. Model Form :

```
#Print Model Summary and goodness of fit
print(Model.summary())

                           Logit Regression Results
==============================================================================
Dep. Variable:             Heart_Risk   No. Observations:                56000
Model:                          Logit   Df Residuals:                    55981
Method:                           MLE   Df Model:                           18
Date:                Mon, 04 May 2026   Pseudo R-squ.:                  0.9674
Time:                        20:20:26   Log-Likelihood:                -1267.3
converged:                       True   LL-Null:                       -38816.
Covariance Type:            nonrobust   LLR p-value:                     0.000
=======================================================================================
                          coef    std err          z      P>|z|      [0.025      0.975]
---------------------------------------------------------------------------------------
const                 -26.1111      0.616    -42.408      0.000     -27.318     -24.904
Chest_Pain              2.7550      0.116     23.743      0.000       2.528       2.982
Shortness_of_Breath     2.6800      0.115     23.298      0.000       2.455       2.905
Fatigue                 2.9332      0.119     24.611      0.000       2.700       3.167
Palpitations            2.7411      0.116     23.576      0.000       2.513       2.969
Dizziness               2.6983      0.116     23.215      0.000       2.470       2.926
Swelling                2.9166      0.118     24.620      0.000       2.684       3.149
Pain_Arms_Jaw_Back      2.7481      0.116     23.703      0.000       2.521       2.975
Cold_Sweats_Nausea      2.8683      0.117     24.421      0.000       2.638       3.099
High_BP                 1.6797      0.111     15.149      0.000       1.462       1.897
High_Cholesterol        1.7543      0.111     15.751      0.000       1.536       1.973
Diabetes                1.6850      0.111     15.183      0.000       1.468       1.903
Smoking                 1.7502      0.111     15.738      0.000       1.532       1.968
Obesity                 1.7070      0.111     15.410      0.000       1.490       1.924
Sedentary_Lifestyle     1.7272      0.111     15.598      0.000       1.510       1.944
Family_History          1.7487      0.111     15.684      0.000       1.530       1.967
Chronic_Stress          1.8116      0.112     16.129      0.000       1.591       2.032
Gender                  1.2532      0.110     11.407      0.000       1.038       1.469
Age                     0.1320      0.005     24.449      0.000       0.121       0.143
=======================================================================================

Possibly complete quasi-separation: A fraction 0.67 of observations can be
perfectly predicted. This might indicate that there is complete
quasi-separation. In this case some parameters will not be identified.
```

3. Model Assumptions :

1. The response is a binary variable Y takes 1 in case of Success [Y=1] and 0 in case of Failure [Y=0].

2. No multicollinearity between explanatory variables using VIF :

□ VIF <= 5: No multicollinearity

- **5 < VIF < 10 : There exists multicollinearity but it's acceptable**

- **VIF >= 10 : High level of multicolinearity**

```
--- VIF Results ---
               Feature        VIF
0                const  13.149802
18                 Age   1.459369
7   Pain_Arms_Jaw_Back   1.453428
8    Cold_Sweats_Nausea  1.449788
5            Dizziness   1.449760
6             Swelling   1.447197
1           Chest_Pain   1.445860
3              Fatigue   1.445207
2  Shortness_of_Breath   1.444974
4         Palpitations   1.441078
10     High_Cholesterol  1.165538
9              High_BP   1.165408
16      Chronic_Stress   1.164450
14  Sedentary_Lifestyle  1.163789
13             Obesity   1.160350
12             Smoking   1.158037
15      Family_History   1.157871
11            Diabetes   1.155911
17              Gender   1.080938
```

4. Model Diagnosis :

To investigate the model's ability to explain the data and its prediction power for the response variable.

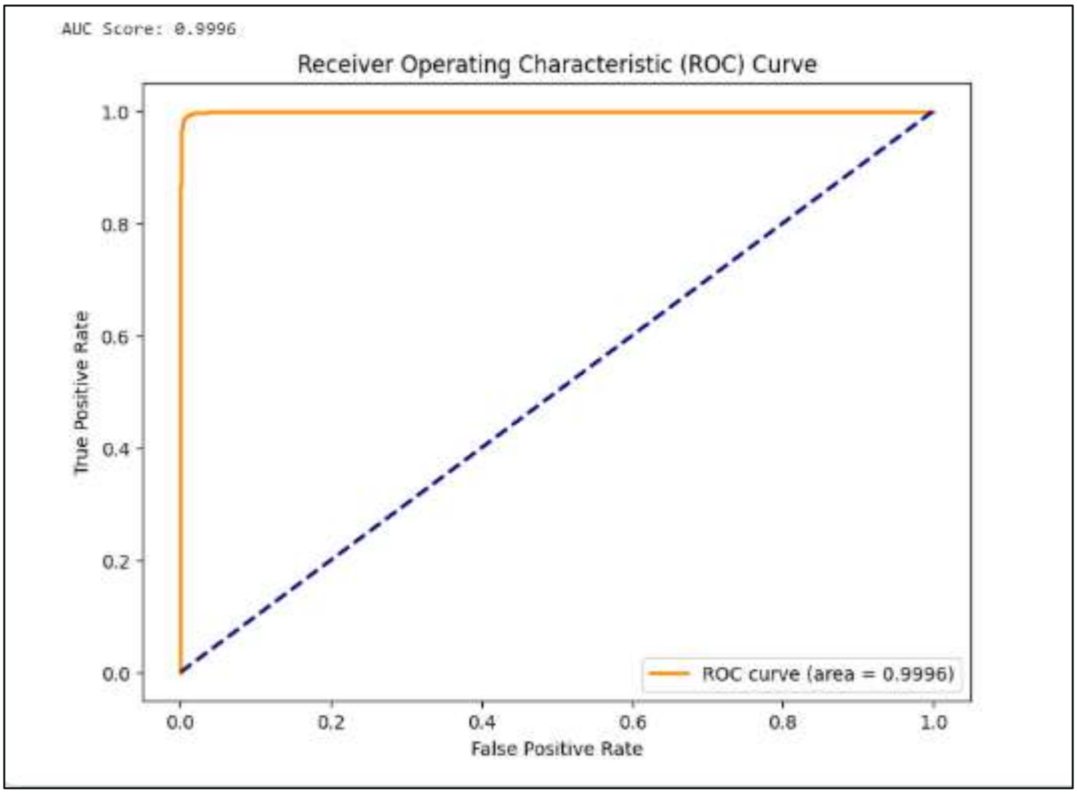
1. Classification table: from the following table the model has 99% correct classification.

```
#4.Check Mpodel Accuracy of predicting with CONFUSION MATRIX
y_prob_test = Model.predict(X_test)
y_pred_test = (y_prob_test > 0.5).astype(int)

print(classification_report(y_test, y_pred_test))
#The Marix Proved That the model has a PERFECT Prediction for the data
```

	precision	recall	f1-score	support
0	0.99	0.99	0.99	6998
1	0.99	0.99	0.99	7002
accuracy			0.99	14000
macro avg	0.99	0.99	0.99	14000
weighted avg	0.99	0.99	0.99	14000

2. The predictive Power of the model represented as the area under the Receiver Operating Characteristic (ROC) Curve, If Area under the curve is higher than 60% the predictive power odf the model is acceptablev.



From the previous chart , AUC = 99% , the model has an excellent predictive power.

Then The model results are reliable for the analysis purpose.